THAPAR INSTITUTE
OF ENGINEERING & TECHNOLOGY
(Deemed to be University)

LAB ASSIGNMENT-4

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2O23

1. Create a class named 'Rectangle' **with two data members – *length* and *breadth* and a function to calculate the area which is 'length*breadth'. The class has three constructors which are:Now, create objects of the 'Rectangle' class having none, one and two parameters and print their areas.**

CODE:

```
#include <iostream>
using namespace std;
class Rectangle {
private:
    float length;
    float breadth;
public:
    Rectangle(float l = 0, float b = 0) {
        length = l;
        breadth = b;
        cout << "Constructor called (Length="
             << length << ", Breadth="
             << breadth << ")" << endl;
    }
```

```
    float area() {
        return length * breadth;
    }
    ~Rectangle() {
        cout << "Destructor called (Length="
             << length << ", Breadth="
             << breadth << ")" << endl;
    }
};
int main() {
    Rectangle r1;
    Rectangle r2(5);
    Rectangle r3(4, 6);
    cout << "\nAreas:" << endl;
    cout << "r1 Area = " << r1.area() << endl;
    cout << "r2 Area = " << r2.area() << endl;
    cout << "r3 Area = " << r3.area() << endl;
    return 0;
}
```

OUTPUT:

```
C:\Users\DELL\Desktop\OOPS

Constructor called (Length=0, Breadth=0)
Constructor called (Length=5, Breadth=0)
Constructor called (Length=4, Breadth=6)

Areas:
r1 Area = 0
r2 Area = 0
r3 Area = 24
Destructor called (Length=4, Breadth=6)
Destructor called (Length=5, Breadth=0)
Destructor called (Length=0, Breadth=0)

--------------------------------
Process exited after 0.1926 seconds with return value 0
Press any key to continue . . .
```

2.Redefine the above program by creating an array of objects of the class Rectangle and calculate area for each object calling different constructors. Also implement constructors with default arguments and destructor in this program

CODE:

```cpp
#include <iostream>
using namespace std;
class Rectangle {
private:
    float length, breadth;
public:
    Rectangle(float l = 0, float b = 0) {
        length = l;
        breadth = b;
        cout << "Constructor called (" << length
             << ", " << breadth << ")" << endl;
    }
    float area() {
        return length * breadth;
    }
    ~Rectangle() {
        cout << "Destructor called (" << length
             << ", " << breadth << ")" << endl;
    }
};
int main() {
    int *pInt = new int;
    *pInt = 10;
    cout << "Dynamic Integer: " << *pInt << endl;
    delete pInt;
    float *pFloat = new float;
    *pFloat = 5.5;
    cout << "Dynamic Float: " << *pFloat << endl;
    delete pFloat;
    int *intArr = new int[3];
    for(int i = 0; i < 3; i++) {
        intArr[i] = i + 1;
    }
    cout << "Dynamic Integer Array: ";
    for(int i = 0; i < 3; i++) {
        cout << intArr[i] << " ";
    }
    cout << endl;
    delete[] intArr;
    float *floatArr = new float[3];
    for(int i = 0; i < 3; i++) {
        floatArr[i] = (i + 1) * 1.1;
    }
    cout << "Dynamic Float Array: ";
    for(int i = 0; i < 3; i++) {
        cout << floatArr[i] << " ";
    }
    cout << endl;
    delete[] floatArr;
    Rectangle *r1 = new Rectangle(4, 5);
    cout << "Area of dynamic object: " << r1->area() << endl;
    delete r1;
    Rectangle *rArr = new Rectangle[2]{
        Rectangle(2, 3),
        Rectangle(6, 7)
    };
    for(int i = 0; i < 2; i++) {
        cout << "Area of object " << i+1
             << ": " << rArr[i].area() << endl;
    }
    delete[] rArr;
    return 0;
}
```

OUTPUT:

```
C:\Users\DELL\Desktop\OOPS

Dynamic Integer: 10
Dynamic Float: 5.5
Dynamic Integer Array: 1 2 3
Dynamic Float Array: 1.1 2.2 3.3
Constructor called (4, 5)
Area of dynamic object: 20
Destructor called (4, 5)
Constructor called (2, 3)
Constructor called (6, 7)
Area of object 1: 6
Area of object 2: 42
Destructor called (6, 7)
Destructor called (2, 3)

--------------------------------
Process exited after 0.1773 seconds with return value 0
Press any key to continue . . .
```

3. Verify the following about destructor **by writing the program:**

(i) Name should begin with tilde sign (~) and must match class name.
(ii) There cannot be more than one destructor in a class.
(iii) Destructors do not allow any parameter.
(iv) They do not have any return type, just like constructors.

When you do not specify any destructor in a class, compiler generates a default destructor and inserts it into your code.

CODE:

```
#include <iostream>
using namespace std;
class Demo {
public:
    Demo() {
        cout << "Constructor called" << endl;
    }
    ~Demo() {
        cout << "Destructor called" << endl;
    }
};
int main() {
    Demo obj;
    return 0;
}
```

OUTPUT:

```
C:\Users\DELL\Desktop\OOPS

Constructor called
Destructor called

--------------------------------
Process exited after 0.1963 seconds with return value 0
Press any key to continue . . .
```

4. Implement dynamic memory allocation. Use new **and** delete **keywords.(For integer variable, float variable, integer array, float array, class objects, Array of Objects)**

CODE:

```cpp
#include <iostream>
using namespace std;
class Sample {
public:
    int x;
    Sample(int value = 0) {
        x = value;
        cout << "Constructor called, x = " << x << endl;
    }
    ~Sample() {
        cout << "Destructor called, x = " << x << endl;
    }
};
int main() {
    int *p = new int;
    *p = 10;
    cout << "Integer value: " << *p << endl;
    delete p;
    cout << endl;
    float *f = new float;
    *f = 5.5;
    cout << "Float value: " << *f << endl;
    delete f;
    cout << endl;

    int *arr = new int[3];
    arr[0] = 1;
    arr[1] = 2;
    arr[2] = 3;
    cout << "Integer Array: ";
    for(int i = 0; i < 3; i++) {
        cout << arr[i] << " ";
```

```
    }
    cout << endl;
    delete[] arr;
    cout << endl;
    float *farr = new float[2];
    farr[0] = 1.1;
    farr[1] = 2.2;
    cout << "Float Array: ";
    for(int i = 0; i < 2; i++) {
        cout << farr[i] << " ";
    }
    cout << endl;
    delete[] farr;
    cout << endl;
    Sample *obj = new Sample(100);
    delete obj;
    cout << endl;
    Sample *objArr = new Sample[2];
    delete[] objArr;
    return 0;
}
```

OUTPUT:

```
C:\Users\DELL\Desktop\OOPS

Integer value: 10

Float value: 5.5

Integer Array: 1 2 3

Float Array: 1.1 2.2

Constructor called, x = 100
Destructor called, x = 100

Constructor called, x = 0
Constructor called, x = 0
Destructor called, x = 0
Destructor called, x = 0

--------------------------------
Process exited after 0.1901 seconds with return value 0
Press any key to continue . . .
```